

Homework 04

Submission Notices:

- Conduct your homework by filling answers into the placeholders given in this file (in Microsoft Word format). Questions are shown in black color, *instructions/hints are shown in italic and blue color*, and *your content should use any color that is different from those*.
- After completing your homework, prepare the file for submission by exporting the Word file (filled with answers) to a PDF file, whose filename follows the following format,
 <StudentID-1>_<StudentID-2>_HW02.pdf (Student IDs are sorted in ascending order)
 E.g., **2252001_2252002_HW02.pdf**
 and then submit the file to Moodle directly **WITHOUT** any kinds of compression (.zip, .rar, .tar, etc.).
- Note that you will get zero credit for any careless mistake, including, but not limited to, the following things.
 1. Wrong file/filename format, e.g., not a pdf file, use “-” instead of “_” for separators, etc.
 2. Disorder format of problems and answers
 3. **Conducted not in English.**
 4. Cheating, i.e., copying other students’ works or let the other student(s) copy your work.

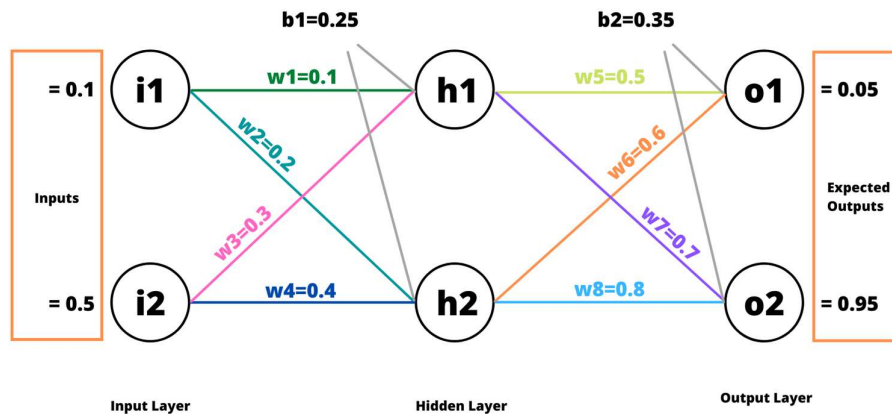
Problem 1. (1.5pts) Column A presents objective measures that are commonly used in decision tree learning, while Column B lists definitions corresponding to the measures in Column A. Match each entry of Column A with the most appropriate definition in Column B.

Column A		Column B	
I	Gini Index	1	measures the statistical significance of the information gain criterion
II	Likelihood-Ratio Chi-Squared Statistics	2	is the splitting criteria of C4.5
III	DKM Criterion	3	measures the divergences between the probability distributions of the target attribute’s values
IV	Gain Ratio	4	requires smaller trees for obtaining a certain error than Gini Index
V	Twoing criteria	5	is the splitting criteria of ID3
VI	Information Gain	6	is the splitting criteria of CART

Please fill your answer in the table below.

I	II	III	IV	V	VI
1	3	4	2	6	5

Problem 2. (3pts) Given a neural network with two inputs, two hidden neurons, two output neurons, as shown aside. Additionally, in the hidden and output layers, each of which will include a **bias** that connects to a constant neuron output of 1. The learning rate is 0.6.



Present all calculations required to perform the backpropagation once (i.e. one forward pass and one backward pass) on the given network in the following cases. (*Precision to 3 decimal places*)

a) Ignore all biases. (1.5pts)

Forward pass (0.5pt)

	h1	h2	o1	o2
Sum	0.160	0.220	0.658	0.768
Sigmoid	0.540	0.555	0.659	0.683

Backward pass (1pt)

w1	w2	w3	w4	w5	w6	w7	w8
-0.001	-0.001	-0.003	-0.004	-0.044	0.019	-0.046	0.019

b) Consider all biases. (1.5pts)

Forward pass (0.5pt)

	h1	h2	o1	o2
Sum	0.410	0.570	0.908	1.118
Sigmoid	0.601	0.639	0.713	0.754

Backward pass (1pt)

w1	w2	w3	w4	w5	w6	w7	w8
-0.001	-0.001	-0.003	-0.005	-0.049	0.013	-0.052	0.014

Problem 3. (2pts) Please write your answer in the following table.

Questions	Answers
1. What's ReLU function and what's the meaning of relu function?	The ReLU (Rectified Linear Unit) function is an activation function used in neural networks that outputs the input directly if it is positive, otherwise, it outputs zero. The meaning behind the ReLU function is to introduce non-linearity to the model, allowing it to learn more complex functions. It's computationally efficient and helps to mitigate the vanishing gradient problem to some extent.
2. What are the disadvantages of sigmoid function? (list at least two)	The disadvantages of the sigmoid function include: Vanishing gradient problem: For inputs with large magnitude, the function saturates at 0 or 1, with a gradient near zero, which significantly slows down the learning. Not zero-centered: Since the sigmoid function outputs values between 0 and 1, the gradients at each layer are either all positive or all negative, leading to zigzagging dynamics during optimization.
3. List names of 3 different types of neural networks?	Three different types of neural networks are: - Convolutional Neural Networks (CNNs), commonly used for image and video recognition. - Recurrent Neural Networks (RNNs), which are used for sequential data like time series or natural language. - Long Short-Term Memory Networks (LSTMs), a special kind of RNNs capable of learning long-term dependencies.
4. What does "stochastic" mean in Stochastic Gradient Descent?	In Stochastic Gradient Descent, "stochastic" refers to the use of a random sample of data points (a mini-batch) to compute the gradient of the loss function, as opposed to using the full dataset for each update. This adds some noise to the optimization process, which can help in escaping local minima and usually leads to faster convergence as compared to using the entire dataset.

Problem 4. (1.5pts) Consider decision trees for classifying these examples.

x_1	1	0	0	0	0	1	0	0	1	1
x_2	0	0	1	0	0	1	1	1	1	1
x_3	0	0	0	0	0	0	1	1	0	0
y	0	0	0	0	0	1	1	1	1	1

a) Do all the decision trees have to be linear classifiers? (0.5pt)

Answer: No

Explanation: They can be used for both linear and non-linear data. The decision tree works on a set of decisions derived from the data and its behavior.

b) Which attribute yields the highest information gain? (0.5pt)

Answer: x_2

Explanation: $IG(x_1)=0.1248$, $IG(x_2)=0.65$, $IG(x_3)=0.2368$

c) You want to flip a single bit in the input of the first data sample (first column), so that learning algorithm of decision tree could not fit the training data. Which bit would you choose to flip? (0.5pt)

Answer: x_2

Explanation:

The algorithm can always fit the data perfectly unless there are two identical examples with different labels. Flipping x_2 to 1 creates such a pair.

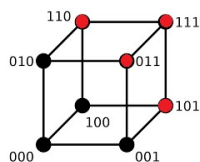
Problem 5. (2pts) Label the following as Y (= yes) or N (= no) depending on whether a linear classifier with a “hard” decision boundary (= a perceptron with a step transfer function) can correctly classify the examples shown. If your answer is Y (= yes), fill in a set of weights that correctly classifies them. Use w_0 as the threshold and w_i as the weight for input x_i . All perceptrons have three Boolean inputs, x_1 , x_2 , and x_3 , and a “dummy” input, x_0 , which is always equal to one. They all compute the decision function $\sum w_i x_i > 0$. You may not transform the input space, i.e., they operate on the stated inputs. Recall that the three Boolean inputs map each possible example to a corner of a three-dimensional cube. *The first one is done for you as an example.*

a) “At least two inputs are 1.”

List the named set S , represented as 3-bit Boolean numbers:

Correctly classifiable?

If yes, weights are $w_0 = -1$; $w_1 = 1$; $w_2 = 1$; $w_3 = 1$



b) (0.5 pts) “Exactly two inputs are 1.”

List the named set S , represented as 3-bit Boolean numbers: $S = \{011, 101, 110\}$

Correctly classifiable?

→ This is not linearly classifiable because there's no linear decision boundary that can separate the points with exactly two 1s from the rest.

c) (0.5 pts) “At most two inputs are 1.”

List the named set S , represented as 3-bit Boolean numbers: $S = \{000, 001, 010, 011, 100, 101, 110\}$

Correctly classifiable?

→ This is linearly classifiable. The weights can be set such that they activate the perceptron when two or fewer inputs are 1

If yes, weights are $w_0 = -1.5$; $w_1 = 1$; $w_2 = 1$; $w_3 = 1$

d) (0.5 pts) “Input $x_1 = 1$, input $x_2 = 0$, input $x_3 = \text{anything}$.”

List the named set S , represented as 3-bit Boolean numbers: $S = \{101, 100\}$

Correctly classifiable?

→ This scenario is linearly classifiable. The weights should ensure that x_1 has a positive influence, x_2 has a negative influence that outweighs x_1 if it is 1 (to prevent activation), and x_3 has no influence.

If yes, weights are $w_0 = -0.5$; $w_1 = 1$; $w_2 = -1$; $w_3 = 0$